

RACK A

MANUFACTURING PACK

Modular CEA growing rack, 4-tier Grow configuration. Everything a fabricator, an assembler and a QC inspector needs to build and accept one unit. Derived from Fusion model CEA_RACK_INTEGRATED_v2 at Rev 5 — the systems-integrated rack, interference-clean. Structural drawings are issued separately as RK-A-DWG Rev 2; validation is recorded in RK-A-QC Rev 2.

INSTALLED	TIERS	MASS, COMPLETE	PARTS IN MODEL	FASTENERS
1456 × 690 × 1960	4 · 16 trays	112.7 kg	45 unique	208 structural
VALIDATION				
26 checks · 35 closed				

01 · SPECIFICATION

FINAL RACK SPECIFICATION

PARAMETER	VALUE	BASIS
Installed envelope	1456 W × 690 D × 1960 H mm	Includes end-mounted electrical enclosure and rear plenum
Envelope, structure only	1256 × 563 × 1960 mm	Frame, decks and trays — the set drawn in RK-A-DWG Rev 2
Envelope, Phase 1 build	1456 × 648 × 1960 mm	Plenum deferred; still set the room out at 690
Structural footprint	1256 × 560 mm	Upright outer faces
Tiers	4 (structure supports 3–5)	Uprights drilled for 5

Bed heights	300 · 700 · 1100 · 1500 mm	Tray support surface
Shelf pitch	400 mm	350 in the 5-tier build
Bed size	1176 × 560 mm	4 × 1020 trays per tier
Tray standard	1020, band 530–545 × 275–285 × 55–70	Design tray 536 × 279 × 64
Tray capacity	16 trays	4 per tier
Rated shelf load	80 kg UDL per tier	Governing service case 38.0 kg
Bed flatness under load	≤ 3 mm; 0.50 mm in service	Flood-and-drain levelness
Irrigation	Flood and drain, closed loop	Flood 22 mm, dwell 10 min. Gravity supply from a terrace reservoir; RK-A-WRS Rev 3
Supply / tray drain / header	DN25 / DN40 / DN50	Drain always exceeds supply; header sized for sequential tier draining
Overflow	30 mm collar moulded into the tray, DN32 line	Sets flood depth mechanically. 11.9 L/min against 7.2 L/min fill
Wet-services corridor	Y 435–525 mm from the front face	Forward of the rear bracing plane, clear of both LED rows
Valve fail-safe	Fill normally CLOSED · drain normally OPEN	A power cut drains every bed and fills none
Flood volume	14.5 L per tier	Freeboard 16.5 L
Lighting	2 bars per tier, 144 mm to canopy	PPFD band 150–210 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$
Airflow	178 m ³ /h per tier, 0.2–0.5 m/s canopy	Linear rear plenum
Electrical	230 V 1-ph, 316 W, 1.37 A, 30 mA Type A RCBO	One feed per rack, local isolator. Type A, not AC — LED drivers give pulsating DC residual current
Floor clearance	150 mm under lowest member	Cleanability
Frame material	Pre-galvanised steel, mill finish	Powder coat optional
Accessory interface	50 mm hole grid, Ø9, two faces	36 positions per upright face

FIFTEEN CHECKS

Every line is a calculation, not an opinion. Two items are not PASS and are called out below the table. **No FEA has been run** — these are closed-form calculations against the as-modelled geometry, which is appropriate for a determinate bolted frame at this scale but is stated plainly so nobody mistakes it for simulation.

#	CHECK	RESULT	CRITERION	VERDICT
1	Static structural load — beam bending at 80 kg UDL	$\sigma = 37.3 \text{ MPa}$, SF 5.6	f_y 210 MPa (YST210)	PASS
2	Shelf deflection, service / fault / design	0.50 / 0.85 / 1.79 mm	$\leq 3 \text{ mm}$ flatness	PASS
3	Upright loading — compression and buckling	2.31 MPa (SF 91); P_{cr} 746 kN (SF 1313)	f_y , Euler at L_e 400 mm	PASS
4	Joint strength — 2 × M8 A2-70 per bracket	shear SF 157, bearing SF 108	98 N per bolt	PASS
5	Rack stability — seismic Zone III	74 N·m vs 489 N·m restoring	IS 1893, A_h 0.050	PASS
6	Tip-over — horizontal pull at top bed	tips at 139 N empty, 307 N loaded	H:D 3.48:1	ANCHOR
7	Water loading — blocked drain	38.0 kg tier; 16.5 L freeboard, 1.8 min	solenoid must close < 60 s	COMMISSION
8	Equipment loading — LED rail with 2 kg fixture	0.93 mm over 1176 mm	visually flat	PASS
9	LED heat clearance	144 mm; air rise 1.0 K at 178 m ³ /h	band 80–200 mm	PASS
10	Fan vibration — beam mode vs fan orders	13.3 Hz vs 33.3 / 233 Hz	no coincidence	PASS
11	Drainage — DN40 tray outlet	30.7 L/min vs 7.2 L/min	drain > fill; 100% fill	PASS

	under 22 mm head	L/min 1111, 4.3×	sequential tiers	
12	Operator access	343 mm opening; 560 mm reach; LI 0.50–0.72	NIOSH LI $\leq$ 1.0	PASS
13	Cleaning access — floor clearance	150 mm (was 90, corrected)	150 mm target	PASS
14	Electrical / water separation	236 mm vertical, IP65, 4 channels	no shared route	PASS
15	Manufacturability — processes and yield	saw, punch, bolt; 2.5–12% waste in 3s	Coimbatore capability	PASS

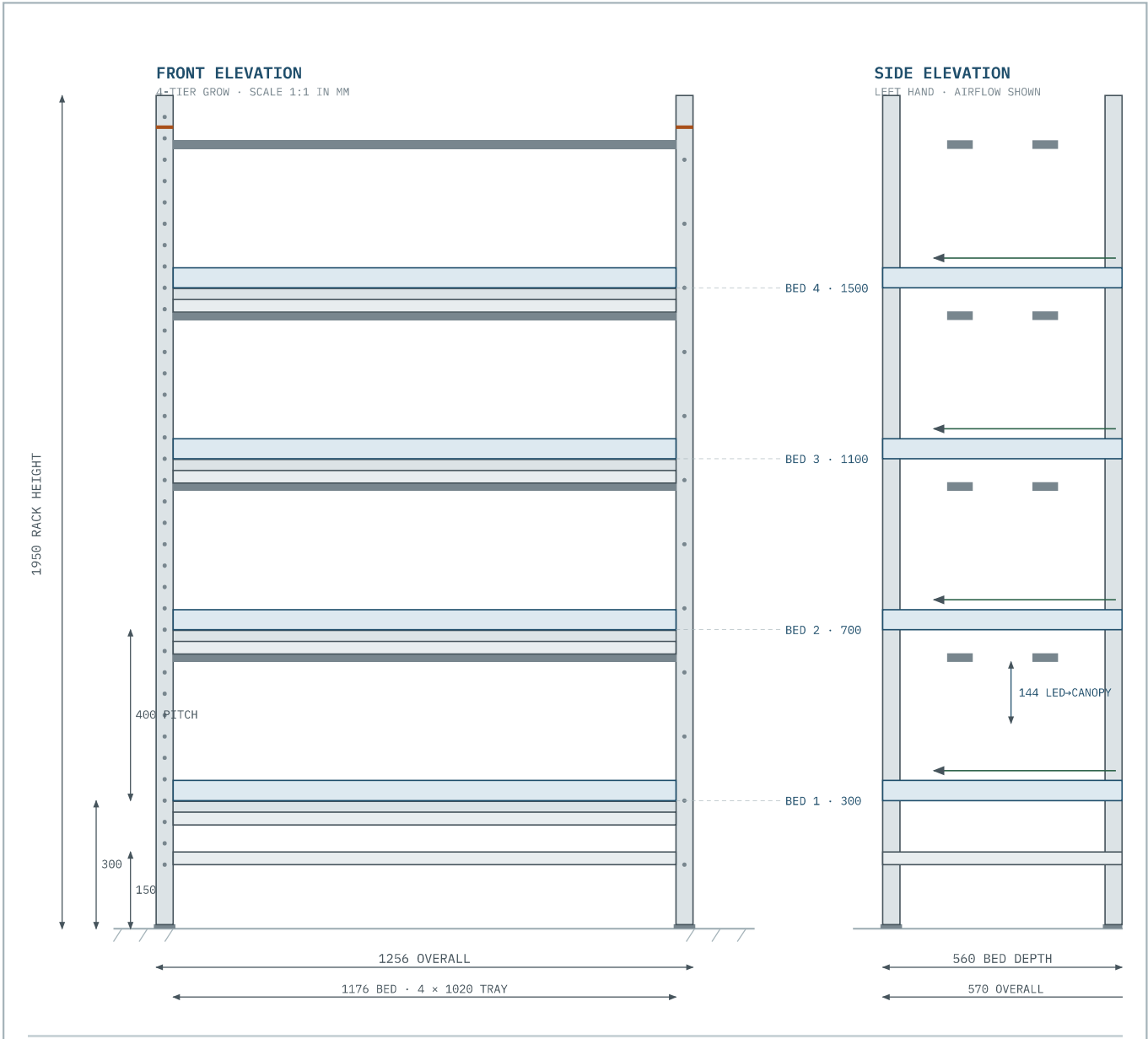
Check 6 — tip-over. Fit the wall anchors. An empty rack tips under a 139 N horizontal pull at the top bed — about 14 kg of force, which a person yanking a stuck tray will exceed without trying. Loaded it takes 307 N. Height-to-depth is 3.48:1, below the 4:1 threshold at which anchoring is normally mandated, so it is defensible as freestanding — but the empty case is genuinely marginal and the fix is trivial. **RK-A-106 wall anchors are included in the BOM and are to be fitted, not optional.** If a rack must stand free, the top tier is to be left unloaded and the rack labelled accordingly.

Check 7 — the overflow interlock is a control specification, not a structural one. If the drain blocks with the supply running, the bed reaches the tray rim in 1.8 minutes. The tray contains it — no water reaches the floor — but only if the solenoid closes. The leak/level sensor and solenoid must be proven to shut off within 60 seconds during commissioning, deliberately, with a blocked drain. Until that test is signed off this is **NEEDS VALIDATION**.

Also still open from the design phase: the depth-plane bracing. Removing the per-tier cross beams and relying on the decks as horizontal diaphragms plus the base frame is sound in principle and the tip-over numbers above assume it holds. It has not been checked by frame analysis. If the prototype shows racking under acceptance test **T16**, the fix is to restore the per-tier cross beams at ₹562.

These fifteen checks are the structural set only. Systems integration added eleven more validation areas and produced **twenty-three Stage 1 findings, all closed** — including four criticals that no structural check could have found: a rear brace dimensioned 452 mm short, an overflow line with a 90 mm break in it, four tiers whose simultaneous drain exceeded the stack, and valve fail-safe polarity that had never been specified. The full register, with corrections and verification method for each, is **RK-A-QC Rev 2**. Read it before quoting this pack.

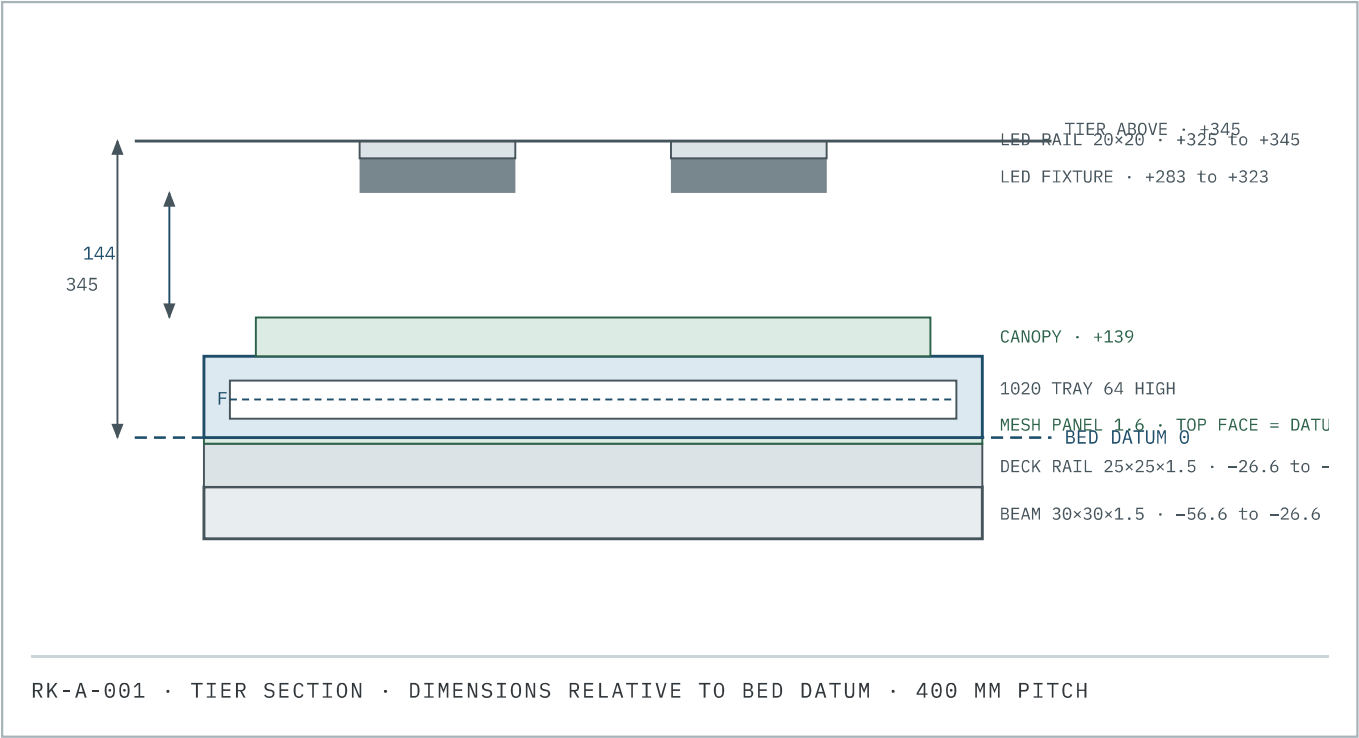
GA DRAWING



RK-A-000 · GENERAL ARRANGEMENT · ALL DIMENSIONS MM · DERIVED FROM THE FUSION MODEL, NOT RE-DRAWN

THE STACK, TIER BY TIER

This is the drawing that matters most on the shop floor, because every vertical dimension in the rack is set by it. Datum is the bed — the tray support surface.



The one rule a fabricator must not break: the mesh panel top face *is* the bed datum. Everything below it (deck rail, beam) hangs from that face, and everything above it (tray, canopy, LED) is measured up from it. Build the deck to the datum, not to the beam.

PART-BY-PART SPECIFICATION

PART NO.	DESCRIPTION	MATERIAL / SECTION	CUT SIZE	QTY	MASS	PROCESS
RK-A-101	Upright	GI SHS 40×40×1.6	1950	4	3.65	Saw, punch 2 faces, deburr, cap
RK-A-102	Beam, long	GI SHS 30×30×1.5	1176	10	1.58	Saw, punch ends, deburr

RK-A-103	Beam, short (base)	GI SHS 30×30×1.5	480	2	0.64	Saw, punch ends, deburr
RK-A-104	Rear X-brace pair	GI flat 25×3	1797 × 2 corrected	1	2.12	Laser or shear, punch, deburr
RK-A-105	Levelling foot	M12 nylon-base adj. foot	Ø50 × 60	4	0.16	Bought out
RK-A-106	Wall anchor bracket	GI plate 4 mm	60 × 60	2	0.15	Laser, bend, slot
RK-A-107	Beam-end L-bracket	GI plate 3 mm	50 × 50 × 40	12	0.07	Laser, bend, punch
RK-A-201	Deck rail, long	GI SHS 25×25×1.5	1176	8	1.30	Saw, punch, deburr
RK-A-202	Deck rail, cross	GI SHS 25×25×1.5	510	16	0.56	Saw, punch, deburr
RK-A-203	Deck mesh panel	GI expanded mesh 1.6, 76% open	1176 × 560	4	1.98	Shear, edge fold, rivet
RK-A-301	LED mounting rail DRAWING HELD	Al 6061 SHS 20×20×1.5	1176	8	0.35	Saw, drill, deburr
RK-A-401	Flood tray, wet module	HDPE 3 mm, natural white	1176 × 560 × 47	4	2.31	Thermoform — collar and both bosses formed in the tool
RK-A-402	Tray drain bulkhead	PP tank connector, EPDM wet-face washer	DN40	4	0.04	Bought out
RK-A-403	Tray strainer, lift-out	SS316 mesh dome	Ø40 × 22	4	0.10	Bought out or press-formed
RK-A-404	Overflow bulkhead	PP tank connector, EPDM wet-face washer	DN32	4	0.03	Bought out
RK-A-501	Plenum face / reflector PHASE 2	White PP 2 mm	1040 × 170	4	0.34	Shear, punch, perforate — pattern set by

the
commissioning
velocity traverse

RK-A-502	Reflective panel, side OPTIONAL	White PP 2 mm	465 × 225	8	0.20	Shear, punch
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UPRIGHT HOLE PATTERN – RK-A-101

This is the accessory interface and the single most important feature in the rack. Get it wrong and nothing else fits.

FEATURE	SPECIFICATION
Hole diameter	Ø9.0 +0.2 / -0.0 (M8 clearance)
Pitch	50.0 ± 0.2 mm cumulative over any 400 mm
First hole	150 mm from the foot end
Count per face	36 (150 to 1900)
Faces	Two adjacent faces, holes on the section centreline
Through condition	Through both walls, in line, single hit
Edge distance	15.5 mm to face edge (min 1.5 × d = 12)
Burr	Both sides, ≤ 0.1 mm – a burr inside the tube is a water trap

Cumulative pitch, not incremental. Measure every hole from the datum end, never hole-to-hole. A 0.2 mm error repeated 36 times is 7 mm at the top, and the four uprights will no longer accept a common beam.

BILL OF MATERIALS

ITEM	PART NO.	QTY	MATERIAL SPECIFICATION	SOURCE
Structural	101-103	16	Pre-galvanised steel hollow section to IS 4923	Local stocked

Structural sections	101–103	10	Pre-galvanised steel hollow section to IS 4726, grade YST210 min, Z275 coating	Local, stocked
Deck sections	201–202	24	Pre-galvanised SHS 25×25×1.5, IS 4923	Local, stocked
Expanded mesh	203	4	GI expanded metal, 1.6 mm, ≥ 70% open area	Local
Flat bar	104	2	GI flat 25 × 3 mm	Local
Plate parts	106–107	14	GI sheet 3 mm and 4 mm	Local laser
Aluminium rail	301	8	Al 6061-T6 SHS 20×20×1.5, mill or clear anodised	Local extruder
Flood tray	401	4	HDPE 3 mm, food-contact grade, natural / white, resin traceable	Thermoformer
Reflective panels	501–502	12	White PP sheet 2 mm, matte, ≥ 85% diffuse reflectance	Local
Levelling feet	105	4	M12 × 60 adjustable, nylon base, ±20 mm	Bought out
Fasteners	—	208	See §07	Bought out

Two material clauses that are not negotiable. The flood tray must be food-contact grade with documented resin traceability — it is the only part in the rack the crop touches. And every cut end of galvanised section exposes bare steel: ends are either capped (uprights, RK-A-101) or sealed with zinc-rich touch-up (all others) before assembly. This is exactly what commercial greenhouse benching does and it is what makes mill-finish GI acceptable in a wet room.

FASTENER SCHEDULE

APPLICATION	FASTENER	QTY	TORQUE
Beam to upright, 24 joints × 2	M8 × 20 hex, SS304 A2-70 + flange nut	48	18 N·m
Deck long rail to beam, 16 ends	M8 × 20 hex, SS304 + flange nut	16	18 N·m
Deck cross to long rail, 32 ends	M8 × 20 hex, SS304 + flange nut	32	18 N·m

Rear brace to upright, 4 ends × 2	M8 × 20 hex, SS304 + flange nut	8	18 N·m
LED rail to grid, 8 rails × 2	M8 × 20 hex, SS304 + flange nut	16	12 N·m
Wall anchor to upright	M8 × 20 hex, SS304 + flange nut	4	18 N·m
Wall anchor to wall	M8 masonry anchor, min 60 mm embed	4	per anchor maker
Levelling feet	M12 × 60 adjustable foot	4	hand + locknut
Deck mesh to deck frame	4.8 mm blind rivet, SS	32	—
Reflective panels to grid	M6 × 16 + penny washer, SS304	48	6 N·m

208 fasteners is high, and it is the honest number. All-stainless was a deliberate call — a rusting bolt in a food room is not worth the ₹330 saved by mixing grades. But the count is a real assembly-time cost, and it is the strongest argument for the v2 tooling: a formed keyhole beam-end removes 48 M8, and a tabbed deck cross rail removes a further 32. That is 38% of the fastener count gone, along with most of the assembly labour.

08 · CUT LIST

NESTED CUT LIST

Order material in batches of three racks. On a single rack, offcut waste runs 22–35%; nested across three it falls to 2.5–12%, and the uprights alone go from 35% waste to 2.5%.

STOCK	CUTS REQUIRED, 1 RACK	BARS, 1 RACK	WASTE	BARS, 3 RACKS	WASTE
GI SHS 40×40×1.6, 6 m	4 × 1950	2	35.0%	4	2.5%
GI SHS 30×30×1.5, 6 m	10 × 1176 · 2 × 480	3	29.3%	7	9.1%
GI SHS 25×25×1.5, 6 m	8 × 1176 · 16 × 510	4	26.8%	10	12.2%
Al SHS 20×20×1.5,	8 × 1176	2	21.6%	5	5.9%

6 m					
GI flat 25×3	2 × 1797	1	40.1%	2	10.2%
GI sheet 3 / 4 mm	12 brackets · 2 anchors	nest	—	nest	—

The brace length changed in Rev 2. RK-A-104 was cut at 1345 mm in Rev 1 and is **1797 mm** — the diagonal of a 1195 × 1342 opening. A 1345 mm brace cannot be fitted at all. It also improves the nest: three racks now waste 10.2% of the flat bar instead of 32.8%.

Saw allowance: add 3 mm kerf per cut when issuing to the shop. The nesting above already assumes it is absorbed in the waste figure. Cut all four uprights from the same bar batch so any mill tolerance is common to the set.

SUBSYSTEM LAYOUTS

Every subsystem below is modelled in CEA_RACK_INTEGRATED_v2 Rev 5 and is interference-clean against the structure. Positions are given from the rack origin: X along the width from the left end, Y from the front face, Z from the floor.

LED mounting

Two RK-A-301 rails per tier at Y 168 and 372, bolted to the **underside of the tier above** — not to their own supports, which is what caused the original 4 mm clash. Only the top tier gets a dedicated rail. Fixtures hang on M8 threaded-rod hangers giving continuous trim; as built, 144 mm to canopy. Drivers remote in the end enclosure, 48 V DC to the tier, IP65 keyed connector.

Ventilation — Phase 2

Folded 0.8 mm GI plenum, 1040 × 120 × 170 at Y 565–685, the fan moving into its end cap and

Ventilation — Phase 1

Standalone EC fans. One 120 × 120 × 38 fan per tier on a grid-mounted bracket at the rear-right of the bed, 178 m³/h. Expect a canopy velocity CV near 33% from a point source.

Traverse five points per tier at commissioning — that measurement is the design input for the Phase 2 plenum, not just an acceptance check.

Irrigation

DN25 riser on the left, X 57–83, fed by gravity from the terrace reservoir. Fill manifold at 7

1000–1300, the fan moving into its end cap and the perforated face RK-A-501 at Y 563–565. Same air, distributed along the full 1176 mm, CV about 10%. Bolts to the same grid holes the Phase 1 bracket uses. Adds 42 mm to installed depth — which is why the room is set out at 690 from the start.

Drainage

All wet services in one corridor at Y 435–525, forward of the rear bracing plane. Each tray drains through a DN40 bulkhead with a lift-out strainer, down to a drain solenoid, across to the DN50 vertical header, and out over a **100 mm air gap** into the tundish at 170 mm. The overflow runs in parallel from the moulded 30 mm collar through its own DN32 bulkhead — two independent penetrations, so a blocked drain still has a path.

from the terrace reservoir. Fill manifold at Z 1000–1300 with all four solenoids grouped at working height. DN20 drop per tier to a nozzle discharging **above the tray rim** — a physical air gap at all sixteen fill points, so back-siphonage is prevented by geometry, not by a check valve. 14.5 L per tier, 10 minute dwell.

Sensors and electrical

T/RH node on the right end face at Z 1200; leak puck in the base pan; header high-level switch at 250 mm. Electrical is all at the **left end**: IP65 enclosure at Z 1700–1900 with the isolator on its front face, cable channel down the left end, per-tier trays. Nothing at mains potential exists below canopy — the tier is 24 V and 48 V only.

FABRICATION INSTRUCTIONS

There is no welding in this rack. Every structural joint is bolted. That is deliberate — it removes the need for pickling and passivation, removes weld distortion from a frame that must hold 3 mm flatness, and lets a general fabrication shop build it. The only welding anywhere is the flood tray, and only if it is fabricated from sheet rather than thermoformed.

01 CUT

All sections to the §08 cut list, cold saw or bandsaw. Square cut, ± 1.0 mm on length, $\pm 0.5^\circ$ on squareness. Do not flame cut galvanised section.

02 PUNCH THE UPRIGHTS

To the RK-A-101 pattern. Cumulative dimensioning from the foot end. Punch or laser; if drilling, use a jig — freehand marking will not hold 0.2 mm cumulative.

03 **PUNCH BEAM AND RAIL ENDS**

Two Ø9 holes per end on the centreline, 25 mm and 75 mm from the end face.

04 **LASER THE PLATE PARTS**

RK-A-106 anchors and RK-A-107 brackets from 3 and 4 mm GI sheet. Bend to $90^\circ \pm 0.5^\circ$.

05 **DEBURR EVERYTHING**

Both faces of every hole, every cut end. A burr inside a tube holds water and starts corrosion from the inside where nobody sees it.

06 **SEAL THE CUT ENDS**

Zinc-rich touch-up on every sawn end. Press plastic caps into both ends of all four uprights. This is what makes mill-finish GI viable in a humid room.

07 **OPTIONAL POWDER COAT**

Grow and Pro tiers only. Polyester powder over the galvanising, RAL to brand. Mask the 50 mm grid holes or ream after coating — a coated hole will not take an M8 cleanly.

08 **DECK ASSEMBLY**

Rivet the expanded mesh into the deck frame, mesh top face flush with the frame top face. This face is the bed datum.

09 **FLOOD TRAY**

Thermoform. **Three features are formed in the tool, not drilled on site:** the 30 mm overflow collar, the DN40 drain boss at 60 mm from the right end and 480 mm from the front edge, and the DN32 overflow boss 60 mm from it on the same line. Form a 2–3 mm fall across the floor toward the drain boss. If fabricated from sheet for the prototype, weld corners inside and out and radius them.

10 **FLOOD-TEST EVERY TRAY**

To the rim, 30 minutes, before it leaves the shop. A leak found in the workshop costs nothing; the same leak on a loaded rack costs a crop.

ASSEMBLY MANUAL

Two people, roughly 90 minutes for a first build, 45 once familiar. Tools: 13 mm spanner and socket, 10 mm spanner, torque wrench, rivet gun, spirit level, tape.

01 LAY OUT THE TWO SIDE FRAMES

On a flat floor, lay one front and one rear upright with the grid faces inward. Bolt the base short beam RK-A-103 between them at 150 mm, using the L-brackets. Repeat for the second side. Do not torque yet.

02 STAND AND CONNECT

Stand both side frames. Bolt the two base long beams RK-A-102 front and rear at 150 mm. The rack is now self-standing but loose. *Second person required.*

03 FIT THE TIER BEAMS

Working bottom to top, bolt a pair of long beams at each tier — beam top face at bed – 26.6 mm, i.e. holes at 250, 650, 1050, 1450 from the floor. Count grid holes from the bottom, do not measure each one.

04 FIT THE REAR X-BRACE

RK-A-104, 1797 mm between hole centres 1757 — check the length against the opening before you cut a batch. Bolted to the rear face of all four upright ends.

SQUARE THE RACK BEFORE TORQUING:

measure both rear diagonals and adjust until they are within 3 mm of each other. The brace locks in whatever squareness you leave it with.

05 TORQUE THE FRAME

All M8 to 18 N·m, working from the base upward. Re-check the diagonals afterwards.

06 FIT THE FEET AND LEVEL

Thread in the four RK-A-105 feet. Level across the width and the depth at the lowest tier.

LEVELNESS IS A FUNCTIONAL REQUIREMENT, NOT A NICETY

— an out-of-level bed gives uneven flood depth across the four trays.

07 FIT THE WALL ANCHORS

RK-A-106 to the rear uprights at 1870 mm, then to the wall. This is not optional — see

validation check 6.

08 DROP IN THE DECKS

One deck assembly per tier, resting on the tier beams, bolted at four corners. Check the deck top face is level and sits at the nominal bed height.

09 FIT THE LED RAILS

Two per tier on the grid at bed + 325 mm. Hang fixtures and set height to 144 mm above expected canopy, or to suit the crop.

10 FIT THE TRAY BULKHEADS

DN40 drain and DN32 overflow tank connectors,

EPDM WASHER ON THE WET FACE

— water pressure then seats the seal instead of lifting it. Drop the lift-out strainer into the drain boss. Do not overtighten; HDPE creeps.

11 PLUMB

Riser and fill manifold on the left, all four fill solenoids grouped at 1100 mm. Drain header on the right in the wet corridor at Y 435–525. Terminate over the tundish with the

100 MM AIR GAP

— never a direct connection.

12 WIRE

Enclosure on the left end at 1700 mm, cable channel down the left end, per-tier trays. Four segregated channels: mains, 48 V LED, 0–10 V dim, RS485 and sensors. Keep every water route below every electrical route.

13 FIT THE FANS

One per tier on the grid bracket at the rear-right of each bed. Phase 2 replaces the bracket with the plenum end cap; the fan and its cable move across unaltered.

14 OPTIONAL PANELS

Side reflectors inboard of the uprights, M6 with penny washers, 6 N·m. The plenum face is a Phase 2 part.

15 FLOOD TEST BEFORE PLANTING

Fill each tier, hold ten minutes, drain. Check for leaks, check flood depth is even across all

Fill each tier, hold ten minutes, drain. Check for leaks, check flood depth is even across all four tray positions, and confirm the drain empties in under a minute.

QC CHECKLIST

INCOMING AND IN-PROCESS

- ☐ **Section grade** — mill certificate for IS 4923 YST210, Z275 coating, retained per batch
- ☐ **Cut length** — every part measured, ± 1.0 mm; sample 100% on the first batch
- ☐ **Upright hole pattern** — cumulative position from datum end, ± 0.2 mm over any 400 mm; check all 4 uprights of every rack
- ☐ **Hole diameter** — $\varnothing 9.0$ go / $\varnothing 9.3$ no-go gauge
- ☐ **Deburr** — visual and finger check both faces of every hole and every cut end
- ☐ **End sealing** — zinc-rich touch-up present on every sawn end; caps fitted to all upright ends
- ☐ **Deck flatness** — mesh top face flush with frame, ≤ 1 mm step
- ☐ **Flood tray** — 30 minute flood test to rim, zero leakage, before despatch
- ☐ **Tray collar crest** — $30 +0/-1$ mm above the tray floor, measured at four points around the crest
- ☐ **Tray boss positions** — centres 60 ± 1.0 mm apart and 480 ± 1.0 mm from the front edge, on all four trays of a rack
- ☐ **Brace length** — RK-A-104 at 1797 ± 1.5 mm, hole centres 1757 ± 1.0 mm **REV 2 CHANGE**

ASSEMBLED RACK — ACCEPTANCE

- ☐ **Squareness** — rear diagonals within 3 mm of each other
- ☐ **Bed height** — each bed within ± 3 mm of nominal 300 / 700 / 1100 / 1500

- ☐ **Bed level** — ≤ 1 mm/m in both axes, every tier, measured loaded if possible
- ☐ **Torque** — sample 20% of M8 joints at 18 N·m, mark each checked bolt
- ☐ **Flood depth uniformity** — flood one tier, measure depth at all four tray positions, spread ≤ 3 mm
- ☐ **Drain time** — full tier empties in under 90 seconds
- ☐ **Drain continuity** — trace tray → strainer → bulkhead → drop → valve → lateral → header, joint by joint; confirm the tundish air gap is 100 mm and is **not** plumbed closed
- ☐ **Fail-safe** — cut power mid-flood: fill solenoids close, drain solenoids open, every bed empties
HOLD POINT
- ☐ **Mass** — assembled dry rack 112.7 ± 7 kg; outside that band the build differs from the model
- ☐ **Leak interlock** — block the drain, confirm solenoid closes within 60 seconds HOLD POINT
- ☐ **Earth continuity** — $< 0.1 \Omega$ from frame to earth terminal
- ☐ **RCD trip** — 30 mA device trips within rated time
- ☐ **Wall anchor** — fitted, both brackets, into sound substrate
- ☐ **Edges** — no sharp edge or burr reachable by hand anywhere on the rack
- ☐ **Labels** — part/serial label and rated load label fitted

PROTOTYPE TEST PLAN

STAGE	OBJECTIVE	TEST	PASS CRITERION
P1	Single deck	Load one deck to 80 kg UDL for 24 h	Deflection ≤ 3 mm, no permanent set after unload
P1	Flood levelness	Flood one bed, measure depth at 4 tray positions	Spread ≤ 3 mm

P2	Single tier live	LED + fan + irrigation for one full crop cycle	Canopy 0.2–0.5 m/s; PPFD 150–210, CV ≤ 15%; no leak
P2	LED thermal	Run 24 h continuous, log fixture and air temperature	Air rise ≤ 3 K; no LED derating
P3	Two-tier frame	Sway test — 300 N horizontal at top bed, front-back and side	Residual deflection ≤ 5 mm; <i>this closes the open depth-plane bracing item</i>
P3	Tip-over	Anchored and unanchored, 300 N pull at top bed	Anchored: no lift. Unanchored: confirm the 139 N / 307 N calculation
P4	Full rack, loaded	All tiers at service load, 30 days	No loosening; re-torque check within 10%
P4	Fault case	Block a drain with supply running	Solenoid closes < 60 s, zero water on floor
P4	Cleaning cycle	Full strip, wash, reassemble, timed	≤ 45 min for one person; no tool needed for deck or tray removal
P5	Production prototype	Full crop trial, both Ooty crops	Yield and quality against the Phase A targets; operator sign-off on handling

This plan is superseded by the eighteen acceptance tests in RK-A-QC Rev 2. Those are numbered T1–T18, carry numeric acceptance criteria and a stated frequency, and two of them exist specifically to close Stage 1 open items: **T16**, the depth-plane sway test, replaces P3 and closes the bracing question; **T10**, the canopy velocity traverse, closes the airflow uniformity item and doubles as the design input for the Phase 2 plenum. The staged P1–P5 sequence above is retained because it is a sensible order in which to build up to a full rack; the acceptance criteria to sign against are the T-numbers.

COSTED BUILD

ELEMENT	DETAIL	COST
GI sections	34.66 kg @ ₹100/kg	₹3,467
Punching	50 mm grid, end prep	₹1,800

L-brackets	12 pcs laser	₹840
Fasteners	208 pcs, SS304	₹1,444
Feet and caps	4 + end caps	₹520
Assembly labour	bolted, no welding	₹1,000
Frame subtotal		₹9,071
Decks × 4	rails 19.44 kg + mesh 7.93 kg + fab	₹3,320
CORE	frame + decks	₹12,391
Flood trays × 4	HDPE, collar and bosses formed in the tool	₹4,835
Tray fittings	4 × DN40 bulkhead, 4 × strainer, 4 × DN32 overflow connector	₹680
LED mounting	8 rails, 16 hangers, 16 saddles, 4 IP65 connector pairs	₹5,460
Irrigation	riser, manifold, 4 fill solenoids, 4 drain solenoids, nozzles, isolation	₹7,650
Drainage	DN50 header, laterals, tundish, air gap, level switch	₹1,300
Ventilation PHASE 1	4 EC fans + grid brackets – plenum deferred	₹2,880
Electrical and control	IP65 enclosure, isolator, Type A RCB0, 24 V PSU, rack controller, segregated cabling	₹9,580
Rack sensors	T/RH node in radiation shield, leak puck, header level switch	₹1,650
Powder coat	optional, over GI	₹2,700
Rev 2 detail changes	moulded collar, grouped fill manifold, level switch	+₹170
GROW – Phase 1 build	as built at Ooty, plenum deferred	₹49,296
Plenum retrofit	Phase 2, month 2-3 – 4 folded GI plenums + perforated faces	+₹2,640
GROW – complete	with ducted plenum	₹51,936
PRO	+ SS304 wet zone, + per-tier sensing	₹56,996

TIER	COST	PRICE @ 40% GM	PRICE @ 45% GM	POSITION
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Core	₹12,391	₹20,650	₹22,500	Entry platform, germination rack
Grow, Phase 1	₹49,296	₹82,160	₹89,630	Flagship as built at Ooty, excludes LED fixtures